

Long Short-Term Memory Networks for Predicting Pore Pressure in Liquefaction

Krishna Kumar & Ellen Rathje

University of Texas at Austin

Overview

- 1 The Liquefaction Problem
- 2 Traditional Approaches Fail
- 3 Sequence Models: RNNs
- 4 LSTM: The Solution
- 5 LSTM for Liquefaction
- 6 Results
- 7 Conclusions

What is Liquefaction?

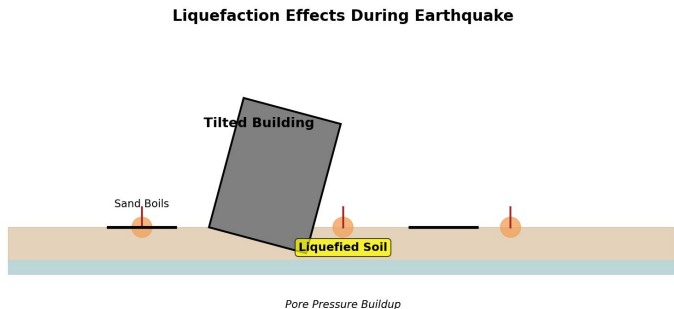


Fig. Liquefaction damage during earthquake

Definition: Saturated soil loses strength during cyclic loading (earthquakes)

Consequences:

- Building settlements and tilting
- Bridge foundation failures

The Physics: Pore Pressure Buildup

Effective Stress Principle:

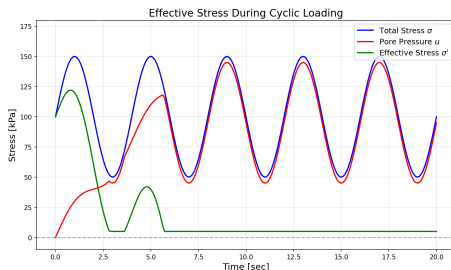
$$\sigma' = \sigma - u \quad (1)$$

where σ' = effective stress, σ = total stress, u = pore water pressure

Pore Pressure Ratio:

$$r_u = \frac{u_{\text{excess}}}{\sigma'_0} \quad (2)$$

Liquefaction occurs when: $r_u \geq 1.0$

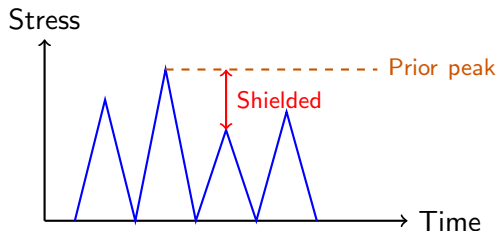


The Challenge: Stress History Dependency

Key Observation: Pore pressure generation depends on **entire stress history**, not just current stress!

The Shielding Effect:

- When current shear stress amplitude $<$ prior peak amplitude
- Pore pressure remains nearly constant ("shielded")
- Sophisticated constitutive models (PM4Sand, PDMY02) **fail** to capture this



Limitations of Constitutive Models

Common Models: PM4Sand, PDMY02, Dafalias-Manzari

Problems:

- 1 Cannot capture shielding effect
- 2 Predict continuous pore pressure increase
- 3 Miss stress history dependencies
- 4 Poor predictions of liquefaction timing
- 5 Require extensive calibration (15-30+ parameters)

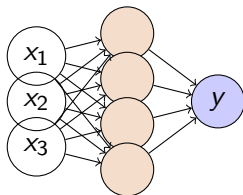
Why Simple Neural Networks Fail Too:

- Take current state as input: $y_t = f(x_t)$
- **No memory** of past events
- Cannot learn temporal dependencies

What We Need: $y_t = f(x_t, x_{t-1}, x_{t-2}, \dots, x_0)$

From Feedforward to Recurrent Networks

Feedforward Neural Network:

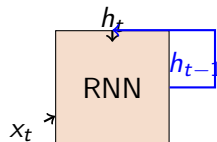


No memory, fixed input size

RNN Equation:

$$h_t = \tanh(W_{xh}x_t + W_{hh}h_{t-1} + b_h) \quad (3)$$

Recurrent Neural Network:



Loops allow information persistence

The Vanishing Gradient Problem

Problem with RNNs:

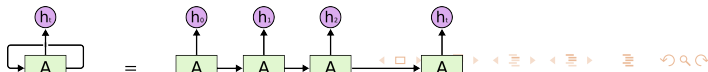
- Gradients multiplied many times during backpropagation
- Gradients vanish ($\rightarrow 0$) or explode ($\rightarrow \infty$)
- Cannot learn long-term dependencies

$$\frac{\partial L}{\partial W} = \sum_{t=1}^T \frac{\partial L}{\partial h_T} \left(\prod_{k=t+1}^T \frac{\partial h_k}{\partial h_{k-1}} \right) \frac{\partial h_t}{\partial W} \quad (4)$$

If $\left| \frac{\partial h_k}{\partial h_{k-1}} \right| < 1$, the inner product decays as $T - t$ grows, so the early steps contribute nothing to the gradient

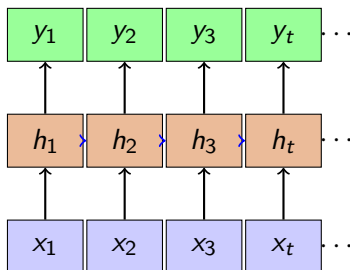
Impact: Can't remember events 10+ steps ago

For liquefaction: the memory horizon here is one window, 800 steps at 0.013s, that is 10.4s. The LSTM state resets at every window.



Sequence-to-Sequence Learning

General Framework: Map input sequence $\rightarrow$ output sequence



For Liquefaction:

- **Input sequence:** Stress history $[\tau_1, \tau_2, \dots, \tau_t]$
- **Output:** Pore pressure ratio $r_u(t)$
- Use sliding window (window = 800 timesteps $\approx$ 2 cycles)

Long Short-Term Memory (LSTM)

Key Innovation: Replace simple RNN cell with complex **memory cell**

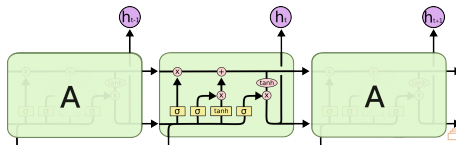
Developed by: Hochreiter & Schmidhuber (1997)

LSTM Can:

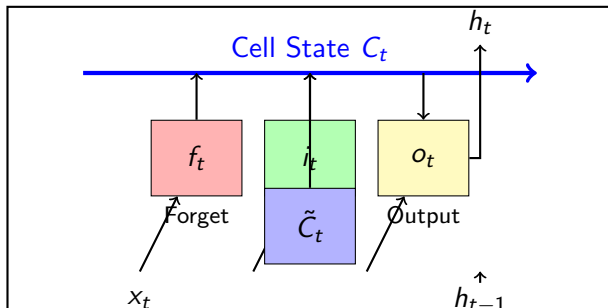
- 1 **Remember** information for long periods
- 2 **Forget** irrelevant information
- 3 **Selectively update** memory

Applications:

- Machine translation
- Speech recognition
- Time series forecasting
- **Now: Geotechnical engineering!**



LSTM Architecture: The Three Gates



Three gates control information flow:

- 1 **Forget gate f_t** : What to remove from memory
- 2 **Input gate i_t** : What new info to store
- 3 **Output gate o_t** : What to output now

LSTM Mathematics: Gate Equations

Input: x_t (current), h_{t-1} (previous hidden), C_{t-1} (previous cell state)

1. Forget Gate (what to discard):

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (5)$$

2. Input Gate (what to add):

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (6)$$

3. Candidate Cell State:

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (7)$$

Where: σ = sigmoid (outputs 0 to 1), $\tanh$ = hyperbolic tangent (outputs -1 to 1)

4. Update Cell State:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (8)$$

- $f_t \odot C_{t-1}$: Forget old information
- $i_t \odot \tilde{C}_t$: Add new information
- $\odot$: Element-wise multiplication

5. Output Gate:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (9)$$

6. Hidden State:

$$h_t = o_t \odot \tanh(C_t) \quad (10)$$

Key Point: Cell state C_t flows with minimal modification $\rightarrow$ solves vanishing gradient!

Why LSTM Solves Vanishing Gradients

Regular RNN:

- Gradient flows through $\tanh$ repeatedly
- After T steps: gradient ≈ 0

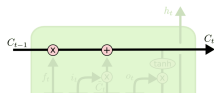
LSTM:

- Cell state provides **highway** for gradients
- Additive updates preserve gradient

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (11)$$

Analogy: LSTM is like a smart notebook

- **Forget gate:** Eraser (remove old notes)
- **Input gate:** Pen filter (decide what to write)
- **Output gate:** Cover (control what readers see)



Problem Formulation

Goal: Predict pore pressure ratio $r_u(t)$ from stress history

Input Features:

- 1 Normalized time: $T_{\text{norm}} = t/600$ s, a fixed constant, so no trial leaks its total duration
- 2 Normalized shear stress: $\tau_{\text{norm}} = \tau/\sigma'_0$
- 3 Relative density: D_r (constant)

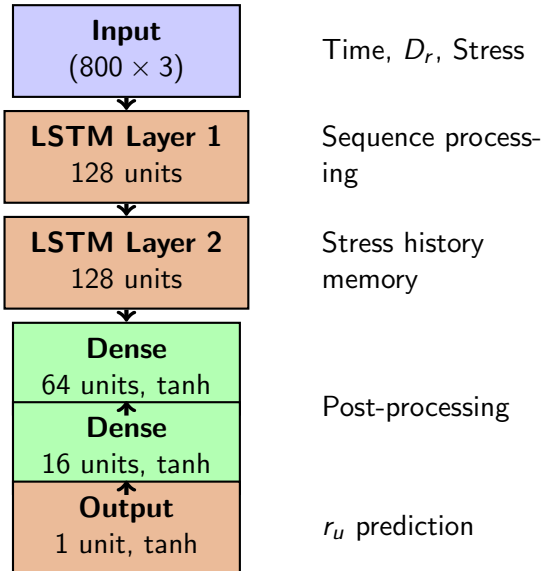
Output: pore pressure ratio $r_u \in [0, 1]$, predicted through a tanh unit whose range is $(-1, 1)$, so the model can go negative and never reaches $r_u = 1$ exactly

Time-Window Approach:

$$\mathbf{X}_t = \begin{bmatrix} T_t & D_r & \tau_t \\ T_{t+1} & D_r & \tau_{t+1} \\ \vdots & \vdots & \vdots \\ T_{t+799} & D_r & \tau_{t+799} \end{bmatrix}_{800 \times 3} \rightarrow \mathbf{Y}_t = r_{u,t+800} \quad (12)$$

Window length: 800 timesteps ≈ 2 loading cycles

Network Architecture



Total Parameters: 200,505

Why LSTM is Perfect for Liquefaction

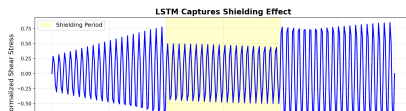
The Shielding Effect - Learned Automatically:

- 1 **Forget gate**: Discards low-stress periods
- 2 **Input gate**: Remembers peak stresses
- 3 **Output gate**: Controls r_u output
- 4 **Cell state**: Long-term stress memory

Learned Behavior:

- When $\tau_{\text{current}} > \tau_{\text{peak}}$:
 - Input gate opens $\rightarrow$ store new peak
 - r_u increases
- When $\tau_{\text{current}} < \tau_{\text{peak}}$:
 - Forget gate filters
 - r_u plateaus (shielding!)

No Hard-Coding! LSTM discovers "if current < peak" logic from data



Experimental Data: Nevada Sand

Data Source: Kwan et al. (2017) — Cyclic Simple Shear Tests

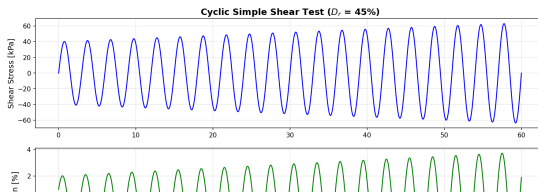
Test Conditions:

- Saturated Nevada sand
- Loose: $D_r = 33\text{-}55\%$ (Exp 9)
- Dense: $D_r = 72\text{-}94\%$ (Exp 8)

Loading Types: Harmonic, Modulated-up, Modulated-down

Training Set:

- 14 trials: 4 from Exp 8, 10 from Exp 9
- 70,631 windows at stride 4
- Validation: 2 whole trials held out before windowing



Training Strategy

Data Preprocessing:

- Augment with 900 static timesteps (initial state)
- Normalize time by a fixed 600 s, stress by σ'_0 , D_r by 100

Memory Optimization:

- Sampling stride = 4 (75% memory reduction)
- File-by-file processing
- Explicit garbage collection
- float32 instead of float64

Hyperparameters:

- Optimizer: Adam, Learning rate: 0.0002
- Batch size: 32, Epochs: 12
- Loss: Mean Squared Error (MSE)
- Early stopping: patience = 1000

Training Performance

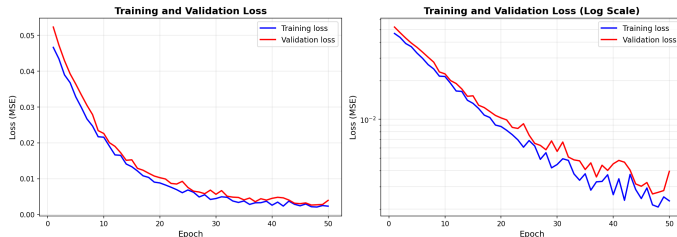


Fig. Training and validation loss over epochs

Observations:

- Rapid initial learning
- Validation is two whole held-out trials, so the gap is a real generalisation gap
- Smooth convergence

Test Set Predictions

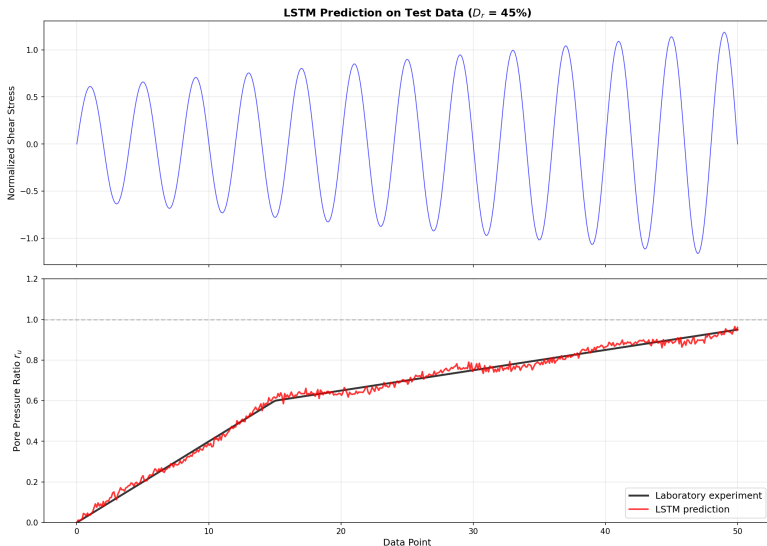


Fig. LSTM predictions vs experimental measurements on test data

Model Performance

Test Set Metrics: pooled over all five held-out trials, 50,687 windows

	Pooled	Exp 8	
MSE	0.116	0.029–0.037	0.0
MAE	0.268	0.123–0.158	0.1
R^2	0.06	0.69–0.72	–2.3

From a 12-epoch run. The model tracks the Experiment-8 modulated-sine trials and fails on the Experiment-9 transient trials.

Twelve epochs underfits.

What LSTM Captures:

- ✓ Shielding effect, on the Experiment-8 trials

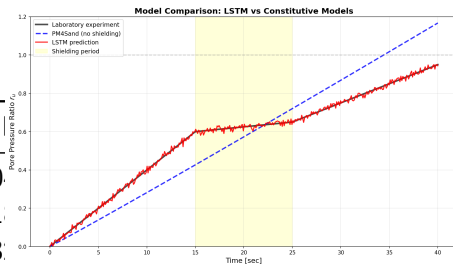


Fig. LSTM vs PM4Sand/PDMY02

vs Constitutive Models:

- PM4Sand: No shielding
- PDMY02: Overestimates r_u
- LSTM: Captures fluctuations

Success Factors:

- 1 Memory mechanism maintains stress history
- 2 Automatic feature learning (gates discover shielding logic)
- 3 Sequence modeling captures temporal context
- 4 Data-driven (no complex parameter calibration)

Advantages:

- ✓ Captures stress history
- ✓ No physics assumptions
- ✓ Fast predictions
- ✓ Generalizes well

Limitations:

- Requires training data
- Less transparent than physics
- Extrapolation uncertain
- Needs GPU for training

Model Improvements:

- Add attention mechanisms
- Incorporate physics constraints
- Multi-task learning (predict strain + r_u)
- Uncertainty quantification

Other Geotechnical Applications:

- Consolidation settlement
- Creep in clays
- Cyclic degradation
- Strain softening

Key Insight: Many geotechnical problems involve history-dependent behavior → LSTM applications!

References

- 1 Choi, Y., and Kumar, K. (2023). "A Machine Learning Approach to Predicting Pore Pressure Response in Liquefiable Sands under Cyclic Loading." *Geo-Congress 2023*.
- 2 Hochreiter, S., and Schmidhuber, J. (1997). "Long Short-Term Memory." *Neural Computation*, 9(8), 1735-1780.
- 3 Kwan, W. S., Sideras, S. S., Kramer, S. L., and El Mohtar, C. (2017). "Experimental Database of Cyclic Simple Shear Tests under Transient Loadings." *Earthquake Spectra*, 33(3), 1219-1239.
- 4 Boulanger, R. W., and Ziotopoulou, K. (2017). "PM4Sand (Version 3.1): A sand plasticity model for earthquake engineering applications."

Code: docs/05-lstm/05a-lstm-liquefaction.ipynb